Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

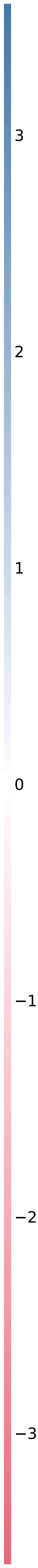
A. Decreased prolactin, decreased FSH, decreased LH

B. Decreased prolactin, increased FSH, increased LH

C. Increased prolactin, decreased FSH, increased LH

D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:



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Finding Reasoning Paths:

1. **Galactorrhea and galactorrheal ductules**: The patient has galactorrhea and galactorrheal ductules, which suggest increased levels of prolactin. Galactorrhea can occur due to elevated prolactin levels from the pituitary gland, typically as a result of a prolactin-secreting tumor (prolactinoma).
2. **Menstrual irregularity and amenorrhea**: The patient has missed her last three periods, consistent with primary amenorrhea or secondary amenorrhea. This could be due to elevated prolactin levels, which suppresses ovulation and hypothalamic function.
3. **Nerve compression and intracranial mass effect**: The presence of galactorrheal ductules, enlarged pituitary stalk, and pituitary edema suggests a macroadenoma, which can cause intracranial mass effect and compress the surrounding tissues.
4. **Neurologic symptoms**: The patient has impaired visual fields, indicating potential involvement of the optic chiasm, which is near the pituitary gland. This supports the possibility of a pituitary tumor.
5. **Serum prolactin levels**: The patientâĢĻs prolactin levels are likely elevated, consistent with a prolactin-secreting pituitary tumor.

Reasoning Process:

1. **Key clinical findings**: The patient presents with galactorrhea, amenorrhea, and visual field changes. These symptoms are commonly associated with a prolactin-secreting pituitary tumor (prolactinoma).
2. **Pathophysiology of a prolactinoma**: A prolactin-secreting pituitary tumor increases prolactin levels, leading to galactorrhea, amenorrhea, and potential optic chiasm compression (visual field changes).

3. **Serum prolactin levels**: Elevated serum prolactin levels are a hallmark of prolactinoma. These levels would typically be significantly higher than the upper normal limit, often in the range of 100-1000 ng/mL or higher, depending on the testing method used.

4. **Neurologic symptoms**: The patientâĢĻs impaired visual fields and enlarged pituitary stalk with pituitary edema suggest compression of the optic chiasm. This is consistent with a macroadenoma, which is a prolactinoma larger than 1 cm.

5. **Differential diagnosis**: Other causes of galactorrhea, such as medications (e.g., metoclopramide, domperidone) or hypothyroidism, are less likely given the absence of a history of medication use and the lack of hypothyroidism-related symptoms (e.g., fatigue, cold intolerance).

6. **Management**: The patient would likely require imaging (e.g., MRI of the pituitary gland) to confirm the diagnosis and tumor size. Prolactin levels would confirm the need for therapy, which may include dopamine agonists (e.g., cabergoline, bromocriptine) to suppress prolactin secretion and tumor growth.

Conclusion:

The patientâĢĻs laboratory results would show **elevated serum prolactin levels**, consistent with a prolactin-secreting pituitary tumor (prolactinoma). This is supported by the clinical findings of galactorrhea, amenorrhea, impaired visual fields, and structural pituitary abnormalities on imaging.

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<answer>

Increased prolactin, decreased FSH, decreased LH

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